

Logs with zeros? Some problems and solutions*

Jiafeng Chen

Harvard Business School

Department of Economics, Harvard University

Jonathan Roth

Department of Economics, Brown University

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Abstract

When studying an outcome Y that is weakly-positive but can equal zero (e.g. earnings), researchers frequently estimate an average treatment effect (ATE) for a “log-like” transformation that behaves like $\log(Y)$ for large Y but is defined at zero (e.g. $\log(1+Y)$, $\operatorname{arcsinh}(Y)$). We argue that ATEs for log-like transformations should not be interpreted as approximating percentage effects, since unlike a percentage, they depend on the units of the outcome. In fact, we show that if the treatment affects the extensive margin, one can obtain a treatment effect of any magnitude simply by re-scaling the units of Y before taking the log-like transformation. This arbitrary unit-dependence arises because an individual-level percentage effect is not well-defined for individuals whose outcome changes from zero to non-zero when receiving treatment, and the units of the outcome implicitly determine how much weight the ATE for a log-like transformation places on the extensive margin. We further establish a trilemma: when the outcome can equal zero, there is no treatment effect parameter that is an average of individual-level treatment effects, unit-invariant, and point-identified. We discuss several alternative approaches that may be sensible in settings with an intensive and extensive margin, including (i) expressing the ATE in levels as a percentage (e.g. using Poisson regression), (ii) explicitly calibrating the value placed on the intensive and extensive margins, and (iii) estimating separate effects for the two margins (e.g. using Lee bounds). We illustrate these approaches in three empirical applications.

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1 Introduction

When the outcome of interest Y is strictly positive, researchers often estimate an average treatment effect (ATE) in logs of the form $E_P[\log(Y(1)) - \log(Y(0))]$, which has the appealing feature that its units approximate percentage changes in the outcome.¹ A practical challenge in many economic settings, however, is that the outcome may sometimes equal zero, and thus the ATE in logs is not well-defined. When this is the case, it is common for researchers to estimate ATEs for alternative transformations of the outcome such as $\log(1+Y)$ or $\operatorname{arcsinh}(Y) = \log(\sqrt{1+Y^2} + Y)$, which behave similarly to $\log(Y)$ for large values of Y but are well-defined at zero. The treatment effects for these alternative transformations are typically interpreted like the ATE in logs, i.e. as (approximate) average percentage effects. For example, among the 11 papers published in the *American Economic Review* since 2018 that interpret a treatment effect for $\operatorname{arcsinh}(Y)$, all but one interpret the result as a percentage effect or elasticity.²

The main point of this paper is that identified ATEs that are well-defined with zero-valued outcomes should not be interpreted as percentage effects, at least if one imposes the logical requirement that a percentage effect does not depend on the baseline units in which the outcome is measured (e.g. dollars, cents, or yuan).

Our first main result shows that if $m(y)$ is a function that behaves like $\log(y)$ for large values of y but is defined at zero, then the ATE for $m(Y)$ will be *arbitrarily sensitive* to the units of Y . Specifically, we consider continuous, increasing functions $m(\cdot)$ that approximate $\log(y)$ for large values of y in the sense that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$. The common $\log(1+y)$ and $\operatorname{arcsinh}(y)$ transformations satisfy this property. We show that if the treatment affects the extensive margin (i.e. $P(Y(1) = 0) \neq P(Y(0) = 0)$), then one can obtain any magnitude for the ATE for $m(Y)$ by rescaling the outcome by some positive factor a . It is therefore inappropriate to interpret the ATE for $m(Y)$ as a percentage effect, since a percentage is inherently a unit-invariant quantity, while the ATE for $m(Y)$ depends arbitrarily on the units of Y .

The intuition for this result is that a “percentage” treatment effect is not well-defined for an individual for whom treatment increases their outcome from zero to a positive value. For example, in our application to Carranza et al. (2022) in Section 5, the treatment induces more people to have positive hours worked. The percentage change in hours is then not well-defined for individuals who would work positive hours under the treatment condition but zero hours under the control condition. Any average treatment effect that is well-defined with zero-valued outcomes must therefore implicitly assign a value for a change along the extensive margin. For logarithm-like transformations $m(\cdot)$, the importance of the extensive margin is determined implicitly by the units of Y . To see why this is the case, consider an individual who works positive hours only if they are treated, so that $Y(1) > 0$ and $Y(0) = 0$. Their treatment effect for the transformed outcome $m(Y)$ is $m(Y(1)) - m(0)$, which becomes larger if the units of Y are re-scaled by some $a > 1$, e.g. if we convert from weekly hours

¹That is, $\log(Y(1)/Y(0)) \approx \frac{Y(1)-Y(0)}{Y(0)}$ when $Y(1)/Y(0) \approx 1$.

²We found 17 papers overall using $\operatorname{arcsinh}(Y)$ as an outcome variable, of which 11 interpret the units; see Appendix Table 1.

if P refers to the sub-population of compliers for an instrument, for instance, then θ is the local average treatment effect (LATE), rather than the ATE in the full population. We are interested in how θ changes if we change the units of Y by a factor of a . That is, how does

$$\theta(a) = E_P[m(aY(1)) - m(aY(0))]$$

depend on a ? Setting $a = 100$, for example, might correspond with a change in units between dollars and cents. Of course, if Y is strictly positive and $m(y) = \log(y)$, then $\theta(a)$ is the ATE in logs and does not depend on the value of a .

We consider “log-like” functions $m(y)$ that are well-defined at zero but behave like $\log(y)$ for large values of y , in the sense that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$. This property is satisfied by $\log(1 + y)$ and $\operatorname{arcsinh}(y)$, for example. Our first main result shows that if the treatment affects the extensive margin, then $|\theta(a)|$ can be made to take any desired value through the appropriate choice of a .

Proposition 1. *Suppose that:*

1. *(The function m is continuous and increasing) $m : [0, \infty) \rightarrow \mathbb{R}$ is a continuous, weakly increasing function.*
2. *(The function m behaves like log for large values) $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$.*
3. *(Treatment affects the extensive margin) $P(Y(1) = 0) \neq P(Y(0) = 0)$.*
4. *(Finite expectations) $E_{P_{Y(d)}}[|\log(Y(d))| \mid Y(d) > 0] < \infty$ for $d = 0, 1$.⁶*

Then, for every $\theta^ \in (0, \infty)$, there exists an $a > 0$ such that $|\theta(a)| = \theta^*$. In particular, $\theta(a)$ is continuous with $\theta(a) \rightarrow 0$ as $a \rightarrow 0$ and $|\theta(a)| \rightarrow \infty$ as $a \rightarrow \infty$.*

[Proposition 1](#) casts serious doubt on the interpretation of ATEs for functions like $\log(1 + Y)$ or $\operatorname{arcsinh}(Y)$ as (approximate) average percentage effects. While a percent (or log point) is entirely invariant to the units of the outcome, [Proposition 1](#) shows that, in sharp contrast, the ATEs for these transformations are *arbitrarily* dependent on units.

2.1 Intuition for [Proposition 1](#)

Loosely speaking, the result in [Proposition 1](#) follows from the fact that a “percentage” treatment effect is not well-defined for individuals who have $Y(0) = 0$ but $Y(1) > 0$.⁷ Any ATE that is well-defined with zero-valued outcomes must implicitly determine how much weight to place on changes along the extensive margin relative to proportional changes along the intensive margin.

When $m(Y)$ behaves like $\log(Y)$ for large values of Y , the importance of the extensive margin is implicitly determined by the units of Y . For intuition, suppose that we re-scale the outcomes so

⁶This assumption simply ensures that $E_{P_{Y(d)}}[|m(aY(d))| \mid Y > 0]$ exists for all values of $a > 0$.

⁷See [Delius and Sterck \(2020\)](#) for an intuitive discussion of this difficulty in the context of the $\operatorname{arcsinh}(\cdot)$ transformation. They write, “the concept of elasticity itself does not make sense with zeros” (p. 21).